

$$y_{i,j} \sim \mathcal{N}(\mu_j, \sigma_e^2) \quad (1)$$

where

$$\mu_j = \alpha_j \quad (2)$$

and σ_e^2 represents random error unrelated to the phylogeny. And the species-specific intercept α_j is elements of the following normal random vectors:

$$\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_n]^T \text{ such that } \boldsymbol{\alpha} \sim N(\mu_{\alpha}, \boldsymbol{\Sigma}_{\alpha}) \quad (3)$$

$$\begin{bmatrix} \sigma^2 & \lambda \times \sigma^2 \times \rho_{12} & \dots & \lambda \times \sigma^2 \times \rho_{1n} \\ \lambda \times \sigma^2 \times \rho_{21} & \sigma^2 & \dots & \lambda \times \sigma^2 \times \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \times \sigma^2 \times \rho_{n1} & \lambda \times \sigma^2 \times \rho_{n2} & \dots & \sigma^2 \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} \sigma^2 & \lambda \times \sigma_{12} & \dots & \lambda \times \sigma_{1n} \\ \lambda \times \sigma_{21} & \sigma^2 & \dots & \lambda \times \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \times \sigma_{n1} & \lambda \times \sigma_{n2} & \dots & \sigma_i^2 \end{bmatrix} \quad (5)$$

Above I have dropped ρ . Now σ^2 is still the Brownian motion rate of evolution across the tree, and σ is the covariance between representative taxa. This is not an uncommon representation but I think we could have been clearer if we presented the last part of our notation as:

$$\boldsymbol{\Sigma} = \sigma_{BM}^2 \cdot \begin{bmatrix} \rho_{1,1} & \lambda \cdots \rho_{1,2} & \dots & \lambda \cdots \rho_{1,n} \\ \lambda \cdots \rho_{2,1} & \rho_{2,2} & \dots & \lambda \cdots \rho_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \cdots \rho_{n,1} & \lambda \cdots \rho_{n,2} & \dots & \rho_{n,n} \end{bmatrix}$$